CONFIDENTIAL



UNIVERSITI TEKNOLOGI MALAYSIA

MID-TERM EXAMINATION

SEMESTER II 2016/2017

SUBJECT CODE	**: SCSI2143**
SUBJECT NAME	**: PROBABILITY & STATISTICAL DATA ANALYSIS**
YEAR/COURSE	**: 1, 2/SCSB, SCSD, SCSJ, SCSR, SCSV**
TIME	**: 8.00 – 10.00 AM**
DATE	**: 14 APRIL 2017**
VENUE	**: BK1, BK3, BK5, BK7, N28**

INSTRUCTIONS TO THE STUDENTS:

Answer all questions in the answer booklet.

NAME	
MATRIC NO.	
SECTION	
LECTURER	

(This question paper consist of 6 pages including this pages)

Question 1 [10 marks]

a) A survey will be given to 100 students randomly selected from the freshmen class at Lincoln High School.

i) What is the population? (1 mark)

ii) What is the sample? (1 mark)

b) State the scale/level of measurement for the following:

i) City of birth (1 mark)

ii) A score on a 5-point quiz measuring knowledge of algebra. (1 mark)

iii) Comparing the number of girls to the number of boys in a classroom. (1 mark)

c) A verbal MUET scores of 20 high school students are recorded as shown below:

280	340	440	490	520	540	560	560	580	580
600	610	630	650	660	680	710	730	740	740

Draw the box-plot of these data. Show the calculations for quartiles and inter-quartile range. (5 marks)

Question 2 [25 marks]

a) Air pollutant index (API) is used as reference to measure the level of air pollution. Table 1 shows the reading of air pollutant index in 19 areas taken at 12 noon. The API readings were grouped into according index.

Table 1: Air pollutant index in 19 areas taken at 12 noon

API	Frequencies
0-50 (good)	1
50-100 (moderate)	2
100-200 (Unhealthy)	6
200-300 (Very Unhealthy)	7
300-500 (Hazardous)	3

i) Sketch a histogram to represent the frequencies of API in table 1. (3 marks)

ii) In order to better understand the air pollutant situation, calculate relative frequencies (2 decimal places) and state the percentage of areas affected with API ≥ 100. (2 marks)

iii) Assume that an API reading of 950 was taken at 12 noon at one of the areas. Does this value affect the mean value for API? Justify your answer. (2 marks)

b) Table 2 shows deaths caused by dengue fever in Malaysia from the year 2006 – 2016.

Table 2: Dengue fever deaths from year 2006 – 2016 in Malaysia

Year	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Deaths	92	98	112	88	134	36	35	92	215	336	237

i) Find mean, mode, median and standard deviation (6 marks)

ii) Number of deaths above 200 is considerably high. Determine the percentile of the deaths 215? (2 marks)

iii) Which statistic measure do you think is the most suitable to show the danger of dengue fever? Explain your answer. (2 marks)

c) Thirty students enrolled in "Statistic Using SPSS" course. The course materials, grades and lecture notes are available to students on e-learning website where frequency of access of each student to any of the pages were recorded. Table 3 below shows frequency of access for the thirty students after one month period.

Table 3: Frequency of access on e-learning of "Statistics Using SPSS" course

Class Interval	Frequencies
0 – 4	1
5 – 9	5
10 – 14	5
15 – 19	6
20 – 24	5

25 – 29	4
30 – 34	2
35 - 39	1
40 - 44	1

i) Find median of the number of access. (4 marks)

ii) Find mode of the number of access. (4 marks)

Question 3 [5 marks]

a) An agricultural cooperative claims that 90% of the watermelons shipped out are ripe and ready to eat. Find the probability that among 18 watermelons shipped out, all 18 are ripe and ready to eat. (1.5 marks)

b) A taxi company in a small town runs a fleet of 20 taxis, which are all aging. Assume that the probability of some type of maintenance being required, for each taxi each day, is 0.2, and that they will or will not require maintenance independently. If Y is the number of taxis that require maintenance on a given day:

i) What is the mean for Y? (1 mark)

ii) What is the variance for Y? (1 mark)

c) An expert shot hits a target 95% of the time. What is the probability that the expert will miss the target for the first time on the fifteenth shot? (1.5 marks)

Question 4 [10 marks]

a) A test has been devised to measure a student's level of motivation during 1st year in university. The motivation scores on this test are approximately normally distributed with mean of 25 and a standard deviation of 6. The higher the score the greater the motivation to do well in university.

i) What percentage of students taking this test will have scores below 10? (1 mark)

ii) What percentage of students taking this test will have scores between 16 and 31? (3 marks)

iii) Aishah is told that 33% of the students taking the test have higher motivation scores that he does? What was Aishah's score? (3 marks)

b) A package delivery service divides their packages into weight classes. Suppose that packages in the weight class of 14 to 20 kg are uniformly distributed, meaning that all weights within that class are equally likely to occur. Let X denote the weight of a package.

i) Write the density function, $f(x)$ of X. (1 mark)

ii) Find the probability that a package weighs at least 15 pounds. (2 marks)

FORMULAS:

$$: \frac{\sum_{i=1}^{N}(x_i-\bar{x})^3}{(N-1)s^3}$$

$$\frac{\sum_{i=1}^{N}(x_i-\bar{x})^4}{(N-1)s^4}$$

$$P(x)=\frac{n!}{x!(n-x)!}p^x(1-p)^{n-x}$$

$$P(x)=(1-p)^{x-1}p$$

$$f(x)=\begin{cases}\frac{1}{b-a}, & a\le x\le b\\ 0 & otherwise\end{cases}$$